MODULE 58 · EXAM STRATEGY - ALL DOMAINS

Full Mock, Part One

Part one of a 75-question mock, weighted across the five domains like the real exam.

PART 1 of 3 QUESTIONS 1-25 SET 6 of 6

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Sitting the mock

- **Allow 60 minutes per part**, 180 for all three, as on the day.
- **Checkpoint**: question 8 by minute 20, question 17 by minute 40.
- **Answer every question** before checking any walkthrough.
- For a true rehearsal, sit all three parts in one session.

> ON THE EXAM
>
> Score each part with the table at its end, then total all 75 after part three.

> An insurer's claim services publish events to a custom EventBridge bus in eu-west-1. If that Region becomes unavailable, producers must keep publishing with no manual action.
>
> Consumers run in eu-west-1 and eu-central-1.

What should the solutions architect implement?

A. A rule forwarding every event to a bus in eu-central-1

B. An SQS queue in front of the bus

C. An EventBridge global endpoint with a secondary bus in eu-central-1 and event replication

D. Producers that retry against eu-central-1 on failure

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Forwarding rule
Fails with the primary Region.

B SQS buffer
Still single-Region.

C Global endpoint
Fails over publishing automatically.

D Client retry logic
Custom failover logic in every producer.

THE SIGNAL

Producers must keep publishing when a Region fails: a global endpoint, not a rule. Module 01.

> A retailer's web tier on EC2 Spot Instances stores shopping carts in server memory and uses sticky sessions. Customers lose carts whenever instances are reclaimed or scaled in.
>
> Traffic is in a single Region.

What should the solutions architect change?

A. Store sessions in Amazon ElastiCache or DynamoDB and make the web tier stateless

B. Increase the sticky session duration

C. Move to On-Demand Instances

D. Disable scale-in

◆ PAUSE AND DECIDE

What should the solutions architect change?

A External session store
Any instance can serve any request.

B Longer stickiness
Sessions still die with the instance.

C On-Demand
Still loses carts on scale-in.

D No scale-in
Wastes money; failures still lose carts.

THE SIGNAL

Stickiness hides statefulness; it does not remove it. Externalise the state. Module 02.

A visual search service on Amazon ECS needs GPU instances of a specific family. The team does not want to patch or manage container hosts.

Fargate does not offer the GPUs required.

What should the solutions architect use?

A ECS on self-managed EC2 Auto Scaling groups

B Amazon ECS Managed Instances

C AWS Batch

D Amazon EKS with self-managed node groups

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Self-managed EC2
The team patches the hosts.

B ECS Managed Instances
Chosen instance types, AWS manages the hosts.

C AWS Batch
Batch jobs; off the in-scope list.

D EKS self-managed
Host management and Kubernetes.

THE SIGNAL

Needs a GPU or a particular family, and nobody wants to patch hosts: ECS Managed Instances. Module 03.

A data provider sells API access in three tiers, each with its own request quota per client key. It also needs AWS WAF protection and response caching.

Clients authenticate with API keys.

Which API option should the solutions architect choose?

A API Gateway HTTP API

B An Application Load Balancer with Lambda targets

C A Network Load Balancer

D API Gateway REST API with usage plans and API keys

◆ PAUSE AND DECIDE

Which API option should the solutions architect choose?

A HTTP API
No usage plans or caching.

B ALB and Lambda
No per-key quotas.

C NLB
No API management at all.

D REST API, usage plans
Per-key quotas, WAF and caching.

THE SIGNAL

Per-client quotas, WAF, caching or a private endpoint rule out HTTP APIs. Module 04.

A news site wants to add a security header and redirect old URLs on every viewer request through CloudFront. The logic is a few lines and needs no network calls.

It runs on millions of requests an hour, and cost matters.

What should the solutions architect use?

A CloudFront Functions on the viewer request

B Lambda@Edge on the origin request

C A Lambda function behind the origin

D AWS WAF rules

◆ PAUSE AND DECIDE

What should the solutions architect use?

A CloudFront Functions
Lightweight, cheapest, sub-millisecond.

B Lambda@Edge
More capable, costlier, not needed.

C Origin Lambda
Every request hits the origin.

D WAF rules
Filters requests; cannot redirect URLs this way.

THE SIGNAL

Default to CloudFront Functions. Move to Lambda@Edge only when a named capability forces it. Module 05.

An insurer has trained its own fraud model with an in-house framework. It must serve real-time predictions with under 100 milliseconds of latency.

Amazon Bedrock does not offer this model.

Where should the solutions architect host it?

A Amazon Bedrock on-demand inference

B Bedrock Knowledge Bases

C A SageMaker AI real-time endpoint

D AWS Lambda calling a Bedrock agent

◆ PAUSE AND DECIDE

Where should the solutions architect host it?

A Bedrock on-demand
Serves Bedrock's models, not this one.

B Knowledge Bases
Retrieval, not model hosting.

C SageMaker AI endpoint
Hosts your own model with low latency.

D Bedrock agent
Wrong service for a custom model.

THE SIGNAL

A model the company built itself, or one Bedrock does not offer: a SageMaker AI endpoint. Module 07.

A retailer has 600 branches that use public DNS resolvers. Security wants central filtering of malicious domains for every branch, plus resolution of private AWS names, without installing appliances at each site.

Branches connect over the internet.

What should the solutions architect recommend?

A. A VPC Resolver inbound endpoint reached by VPN from every branch

B. Route 53 Global Resolver with its domain filtering

C. A DNS appliance at each branch

D. Public hosted zones for every private name

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Inbound endpoint via VPN

600 VPNs to run.

B Global Resolver

Internet-reachable resolver with central filtering.

C Appliance per site

600 appliances.

D Public zones

Exposes private names.

THE SIGNAL

Hundreds of branches, central DNS filtering, no appliance per site: Global Resolver. Module 09.

> Pods in an EKS cluster need IAM permissions. Some pods run on Fargate, and the rest on EC2 nodes.
>
> The team wants one mechanism that works for every pod.

Which mechanism should the solutions architect choose?

A EKS Pod Identity

B Node instance profiles

C Access keys in Kubernetes secrets

D IAM roles for service accounts

◆ PAUSE AND DECIDE

Which mechanism should the solutions architect choose?

A Pod Identity

Not supported on Fargate pods.

B Instance profiles

Every pod on a node gets the same rights.

C Access keys

Long-term credentials.

D IRSA

Works on both Fargate and EC2 pods.

THE SIGNAL

Pod Identity whenever supported. Fargate pods, Windows nodes and EKS Anywhere still need IRSA. Module 15.

A data team in account A runs a job that copies objects from its own bucket into a bucket in account B. Each copy is a single operation.

Assuming a role in account B makes the job lose access to account A's bucket.

What should the solutions architect do?

A Assume the role in account B, then assume back

B Keep the job's account A role, and add a bucket policy in account B granting that role write access

C Copy objects through a local download

D Make account B's bucket public for writes

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Chain roles

One call cannot hold both sets of permissions.

B Resource policy in B

One identity, access to both buckets.

C Download and upload

Slow and costly.

D Public bucket

Unsafe.

THE SIGNAL

Moving data between your resource and theirs in one call needs a resource policy, because a role swaps permissions. Module 16.

A manufacturer's on-premises servers call AWS APIs using long-term access keys stored in configuration files. Each server already has an X.509 certificate from the company's certificate authority.

Security wants the access keys removed.

What should the solutions architect implement?

A IAM Roles Anywhere, trusting the company CA, with profiles mapping to IAM roles

B Rotate the access keys monthly

C Systems Manager hybrid activations only

D IAM Identity Center users for each server

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Roles Anywhere**
Temporary credentials from existing certificates.

B **Monthly rotation**
Still long-term keys.

C **Hybrid activations**
For managing servers, not app credentials.

D **Identity Center**
For people, not workloads.

THE SIGNAL

Match the mechanism to what the workload already has: a certificate means Roles Anywhere. Module 17.

A European public-sector body requires that its cloud is operated only by EU-resident staff, with infrastructure and metadata independent of other AWS Regions.

Data residency alone is not sufficient.

What should the solutions architect recommend?

A Any EU Region with an SCP limiting Regions

B An EU Region with customer managed KMS keys

C The AWS European Sovereign Cloud

D AWS Outposts in the body's data centre

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A EU Region and SCP
Residency only.

B EU Region and CMKs
Residency and key control only.

C European Sovereign Cloud
Operated in the EU by EU residents, independent.

D Outposts
Still managed from a parent Region.

THE SIGNAL

Residency is solved with Regions and policy. Sovereignty may need a different operating model. Module 23.

A company must collect CloudTrail, VPC Flow Logs and security findings from 120 accounts into one store. Analysts query a year of data, and partner tools expect the OCSF schema.

The team wants the least custom pipeline work.

What should the solutions architect use?

A Custom Firehose streams and Glue jobs

B CloudTrail Lake

C CloudWatch Logs in each account

D Amazon Security Lake with a delegated administrator

◆ PAUSE AND DECIDE

What should the solutions architect use?

A **Custom pipeline**
Builds what Security Lake provides.

B **CloudTrail Lake**
Closed to new customers; CloudTrail only.

C **Per-account logs**
Not central, no OCSF.

D **Security Lake**
Central, OCSF-normalised, organisation-wide.

THE SIGNAL

Organisation-wide security data in OCSF with least effort: Security Lake. Module 21.

A bank's customer assistant on Amazon Bedrock sometimes repeats customers' card numbers and phone numbers in its responses.

Card numbers must be blocked, and phone numbers masked.

What should the solutions architect configure?

A. A denied topic for payments

B. A Bedrock guardrail with sensitive information filters, blocking card numbers and masking phone numbers

C. A larger model

D. Contextual grounding checks

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Denied topic

Blocks whole conversations about payments.

B Sensitive information filters

Per-entity block or mask actions.

C Larger model

No guarantee.

D Grounding checks

Checks support by sources, not PII.

THE SIGNAL

Personal data in prompts or responses: sensitive information filters, block or mask. Module 24.

A company runs the same EC2 instance family in one Region all year, at a steady 80 dollars an hour. It has no plans to change family or Region for three years.

It wants the largest commitment discount.

What should the solutions architect recommend?

A EC2 Instance Savings Plans

B Compute Savings Plans

C Spot Instances

D On-Demand Capacity Reservations

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A EC2 Instance SP

Largest Savings Plan discount for a fixed family and Region.

B Compute SP

More flexible, smaller discount.

C Spot

Interruptible.

D Capacity reservations

Capacity, not a discount.

THE SIGNAL

Fixed family and Region, maximum discount: EC2 Instance Savings Plans. Module 25.

A publisher summarises two million archived articles with Amazon Bedrock each night. Results are needed by the next morning.

The team wants the lowest inference cost.

What should the solutions architect use?

A On-demand inference with more parallel calls

B Provisioned Throughput

C A larger model

D Bedrock batch inference

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Parallel on-demand

Full on-demand price.

B Provisioned Throughput

Pays for reserved capacity.

C Larger model

Costs more per token.

D Batch inference

Lower price for asynchronous jobs.

THE SIGNAL

Large, non-urgent jobs: batch inference at a lower price than on-demand. Module 27.

A DynamoDB table serves steady, predictable traffic of about 2,000 writes per second all day. It currently uses on-demand capacity.

Traffic has not varied by more than 10 percent in a year.

How should the solutions architect reduce cost?

A Keep on-demand and add DAX

B Move to the Standard-Infrequent Access table class

C Switch to provisioned capacity with auto scaling, and buy reserved capacity

D Add global secondary indexes

◆ PAUSE AND DECIDE

How should the solutions architect reduce cost?

A On-demand plus DAX
Caches reads; writes dominate.

B Standard-IA class
Saves on storage, not throughput.

C Provisioned plus reserved
Cheapest for steady, predictable load.

D More indexes
Adds cost.

THE SIGNAL

Steady, predictable traffic: provisioned capacity, plus reserved capacity. Module 31.

> An application writes 5 TB of logs a month to S3. They are never read after 30 days, must be kept for a year for audit, and retrieval within 12 hours is acceptable.
>
> The company wants the lowest storage cost.

What should the solutions architect configure?

A S3 Standard for a year

B A lifecycle rule moving objects to S3 Glacier Deep Archive after 30 days and expiring them after a year

C S3 Intelligent-Tiering

D S3 One Zone-IA

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Standard
Most expensive.

B Lifecycle to Deep Archive
Cheapest, retrieval within 12 hours.

C Intelligent-Tiering
Monitoring fees for data with a known pattern.

D One Zone-IA
Costlier than Deep Archive.

THE SIGNAL

Known access pattern, retrieval within a day: lifecycle rules to Deep Archive. Module 31.

> A company is moving a 3 TB MySQL database from on premises to Aurora MySQL. Downtime must be under ten minutes.
>
> There is a 1 Gbps Direct Connect link.

What should the solutions architect do?

A AWS DMS full load plus ongoing replication, then cut over

B DMS Schema Conversion first, then a full load

C mysqldump during a weekend outage

D AWS Transform MGN for the database server

◆ PAUSE AND DECIDE

What should the solutions architect do?

A DMS full load plus CDC

Same engine; sync, then a brief cutover.

B Schema Conversion first

No conversion needed for the same engine.

C mysqldump

Hours of downtime.

D Rehost the server

Moves a server, not to Aurora.

THE SIGNAL

Same engine means no conversion step. Full load plus change data capture keeps downtime short. Module 34.

> A company has 70 ASP.NET Framework applications on Windows servers. It wants to run them on Linux to cut licence costs.
>
> It wants as much of the code porting automated as possible.

What should the solutions architect recommend?

A Rehost them to Windows on EC2

B Rewrite them in Java

C Run them on Windows containers on ECS

D AWS Transform for .NET to port them to cross-platform .NET on Linux

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Rehost to Windows
Keeps the Windows licences.

B Rewrite in Java
Manual and slow.

C Windows containers
Still Windows licences.

D AWS Transform for .NET
Automated porting to Linux.

THE SIGNAL

Windows .NET Framework to Linux, automated: AWS Transform for .NET. Module 35.

> A company needs disaster recovery for an order system with an RPO of five minutes and an RTO of two hours. Cost while idle must be low.
>
> The infrastructure is defined in CloudFormation.

Which DR strategy should the solutions architect choose?

A. Backup and restore with daily snapshots

B. Pilot light: replicate the database continuously, keep core services minimal, and deploy the rest from templates on failover

C. Warm standby at full size

D. Multi-site active-active

◆ PAUSE AND DECIDE

Which DR strategy should the solutions architect choose?

A Daily backups
RPO of a day.

B Pilot light
Minutes of RPO, hours of RTO, low idle cost.

C Full warm standby
Costs far more than needed.

D Active-active
Most expensive option.

THE SIGNAL

Match the strategy to the numbers: RPO minutes, RTO hours, low idle cost means pilot light. Module 37.

> A group with 90 accounts must back up every production EBS volume and RDS database daily, keeping copies for 35 days. This must include resources created in future.
>
> Teams must not be able to opt out.

What should the solutions architect implement?

A. A Lambda function that snapshots tagged resources

B. A backup plan created by each team

C. AWS Organizations backup policies applying an AWS Backup plan to resources by tag

D. Daily EBS snapshots with Data Lifecycle Manager in each account

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Custom Lambda
Custom code to maintain.

B Team-owned plans
Teams can opt out.

C Organisation backup policies
Central, enforced, covers new resources.

D DLM per account
EBS only; managed per account.

THE SIGNAL

Enforced backups across an organisation: backup policies in AWS Organizations. Module 39.

> A platform team wants encryption, versioning and a cost-centre tag applied automatically to every S3 bucket in any CDK app. This must happen before synthesis.
>
> Teams should not need to write this themselves.

What should the solutions architect provide?

A. A CDK aspect in a shared library

B. A Config rule with remediation

C. An SCP

D. A code review checklist

◆ PAUSE AND DECIDE

What should the solutions architect provide?

A CDK aspect

Applies settings to every construct before synthesis.

B Config rule

Acts after deployment.

C SCP

Denies; does not set properties.

D Checklist

Manual.

THE SIGNAL

Fix or flag every construct in CDK apps before synthesis: an aspect. Module 41.

> A team builds a container image separately in its test, staging and production accounts. A production fault came from a dependency version that differed from what staging tested.
>
> Production must run exactly what passed staging.

What should the solutions architect recommend?

A Run the tests again in production

B Pin more dependency versions

C Build in production only

D Build the image once in a tooling account and promote the same image digest to every account

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Retest in production

Tests a different build.

B Pin versions

Narrows the gap; still rebuilds.

C Build in production

Untested build ships.

D Build once, promote

Tested artefact is the shipped artefact.

THE SIGNAL

Production must run exactly what was tested: build once and promote the artefact. Module 42.

> A team's pipeline uses queued execution mode. During busy days, dozens of commits queue up and each deploys in turn, delaying the latest fix by hours.
>
> Only the newest commit needs to reach production.

What should the solutions architect change?

A Switch to parallel execution mode

B Switch to superseded execution mode

C Add more stages

D Add a manual approval to skip old runs

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Parallel

Runs them all at once; no stage rollback.

B Superseded

Newer runs replace waiting ones.

C More stages

Slower.

D Manual approval

Manual work for a built-in behaviour.

THE SIGNAL

Rapid commits, deploy only the latest: superseded execution mode. Module 42.

> A Lambda function must publish custom metrics for order value and payment provider with several dimensions. Calling the CloudWatch API on each invocation added latency.
>
> The team wants no synchronous metric calls.

What should the solutions architect recommend?

A Batch PutMetricData calls in a background thread

B X-Ray annotations

C CloudWatch embedded metric format in structured log lines

D Write metrics to DynamoDB

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Background thread

Lambda may freeze before the thread sends.

B X-Ray annotations

Index traces; not metrics.

C Embedded metric format

Metrics extracted from logs asynchronously.

D DynamoDB

A metrics system built by hand.

THE SIGNAL

Custom metrics from Lambda without API calls: embedded metric format. Module 45.

Part one: score and next steps

RESULT ON THESE 25	WHAT TO DO
20 or more	On track. Continue to part two
16 to 19	Passing range but thin. Review misses first
12 to 15	Rewatch the modules named in your misses
Below 12	Return to the domain papers for weak domains
Guessed but correct	Count as a miss when deciding
Over 60 minutes	Practise pacing with the checkpoints

ON THE EXAM

Scaled scoring means a raw count cannot be converted to 720. Treat these bands as a guide, and aim well clear.